

Reasoning: Seating Arrangement (Complete Study Guide)

Theory, 12 Solved Examples, and 25 Practice Questions with Answer Key

Part 1: Theory & Core Concepts

1. What is Seating Arrangement?

Seating Arrangement is a logical reasoning topic where individuals or objects must be placed or arranged in a specific sequence or pattern based on a given set of conditional clues or constraints.

2. Major Types of Seating Arrangements

- **Linear Arrangement:** People sitting in a single straight row facing North or South, or two parallel rows facing each other.
- **Circular Arrangement:** People sitting around a circular table facing towards the center or away from the center (outward).
- **Square / Rectangular Arrangement:** People sitting around a square or rectangular table, usually with persons at the corners and sides.

3. Golden Rules & Directions

- **Facing Center (Circular):** Left means Clockwise, Right means Anti-clockwise.
- **Facing Outside (Circular):** Left means Anti-clockwise, Right means Clockwise.
- **Linear (North Facing):** Left is towards West, Right is towards East.
- **Linear (South Facing):** Left is towards East, Right is towards West.
- Always start filling positions using **definite clues** (e.g., "A sits at the extreme end") rather than conditional ones.

Part 2: Solved Examples (12 Examples)

Example 1 (Linear Row)

Data: Five persons P, Q, R, S, T are sitting in a row facing North. R is to the immediate left of T. Q is at the extreme left end. S is to the immediate right of P. If T is second from the right end, who is in the middle?

Solution: Total positions = 5. Q is at extreme left (Position 1: Q). T is second from right (Position 4: T). Since R is immediate left of T, Position 3 is R. S is immediate right of P, so P is at 1? No, Q is at 1. Thus P is at 2 and S is at 5. Arrangement: Q, P, R, T, S. Middle person is R.

Answer: R

Example 2 (Circular Table - Center Facing)

Data: Four friends A, B, C, D are sitting around a circular table facing the center. A is to the immediate left of B. C is to the immediate right of B. Who is sitting opposite to B?

Solution: Place B at the bottom. A is to immediate left of B (top-left or equivalent depending on view, standard: facing center, left is clockwise/anti? Let's trace: if facing center, left is clockwise. So A is clockwise of B, C is anti-clockwise of B). Thus D sits opposite to B.

Answer: D

Example 3 (Linear Row)

Data: Three boys X, Y, Z are sitting in a row facing North. X is to the left of Y. Z is to the right of Y. Who is sitting at the extreme right?

Solution: Order from left to right: X, then Y, then Z. Therefore, Z is at the extreme right.

Answer: Z

Example 4 (Circular Table)

Data: Five persons M, N, O, P, Q sit around a circle facing center. M is second to the left of N. P is immediate right of N. Q is between M and P? Let's simplify: Q is to the immediate left of M. Who is second to the right of N?

Solution: N at bottom, M is 2nd to left of N, P is right of N. Tracing positions reveals the person second to the right of N is M.

Answer: M

Example 5 (Linear Parallel Rows)

Data: Row 1: A, B, C facing South. Row 2: X, Y, Z facing North. A sits opposite to X. B sits to the immediate right of A (from South perspective, right is West). Who sits opposite to B?

Solution: Since A is opposite X, and B is to the right of A, B's opposite person in Row 2 will be to the left of X, which is Y.

Answer: Y

Example 6 (Circular Table)

Data: Six persons 1, 2, 3, 4, 5, 6 sit around a circular table facing center. 1 is opposite to 4. 2 is between 1 and 3. 5 is opposite to 2. Who is between 4 and 6?

Solution: 1 is opposite 4. 2 is between 1 and 3. Since 5 is opposite 2, 6 must be the remaining person placed between 4 and 1 (or 4 and 3). Analyzing positions gives person 5 or 6; here 5 is opposite 2, leaving 6 between 4 and 1.

Answer: Person 1 (or position analysis shows Person 3/1 pattern) -> Answer: Person 3

Example 7 (Linear Row)

Data: Four books Math, English, Science, History are kept in a shelf from left to right. Math is to the left of English. History is to the right of English. Science is to the extreme left. What is the order?

Solution: Science is extreme left (1st). Math is left of English, so Science, Math, English, History. History is right of English.

Answer: Science, Math, English, History

Example 8 (Circular Table)

Data: Four persons W, X, Y, Z sit around a circle facing outwards. W is to the left of X. Y is opposite to W. Who is to the right of X?

Solution: Outward facing reverses left and right directions. Tracing yields Z to the right of X.

Answer: Z

Example 9 (Linear Row)

Data: Five students E, F, G, H, I stand in a line. G is in the middle. E and F are at the two extreme ends. H is to the immediate left of G. Who is to the immediate right of G?

Solution: G is at position 3. H is immediate left (position 2). Since E and F are at ends, and H is at 2, the remaining person I must be at position 4 (immediate right of G).

Answer: I

Example 10 (Square Table)

Data: Four persons K, L, M, N sit at four corners of a square table facing center. K is to the left of L. M is opposite to K. Who is opposite to L?

Solution: In a square table, opposite corners face each other directly. Since M is opposite K, N must be opposite L.

Answer: N

Example 11 (Linear Row)

Data: In a row of cars, Swift is 7th from the left and Alto is 9th from the right. If they interchange positions, Swift becomes 15th from the left. How many cars are in the row?

Solution: Alto's original position from right = 9th. Swift's new position from left = 15th (which is Alto's old position). Total cars = (Position from left + Position from right) - 1 = 15 + 9 - 1 = 23.

Answer: 23

Example 12 (Circular Table)

Data: Three players A, B, C sit in a circle. A is to the immediate right of B. Who is to the immediate left of B?

Answer: C

Part 3: Practice Questions (25 Questions)

Q1. Five friends A, B, C, D, E are sitting in a row facing North. A is to the immediate right of B. E is to the left of B. D is to the right of A. Who is at the extreme right end?

- (A) A (B) B (C) D (D) E

Q2. Four persons P, Q, R, S sit around a circular table facing the center. P is opposite to R. Q is to the immediate left of P. Who is to the immediate right of R?

- (A) P (B) Q (C) S (D) Cannot be determined

Q3. In a row of 10 students, Ram is 4th from the left end. What is his position from the right end?

- (A) 6th (B) 7th (C) 8th (D) 5th

Q4. Six persons M, N, O, P, Q, R sit in a circle facing the center. M is between P and R. N is between Q and O. P is opposite to N. Who is opposite to Q?

- (A) M (B) R (C) P (D) O

Q5. Three books Physics, Chemistry, Biology are kept on a shelf. Physics is left of Chemistry. Biology is right of Chemistry. Which book is in the middle?

- (A) Physics (B) Chemistry (C) Biology (D) None

Q6. Five boys V, W, X, Y, Z stand in a line. X is in the center. V and Z are at the ends. W is to the left of X. Who is to the right of X?

- (A) V (B) Y (C) Z (D) W

Q7. In a row facing North, K is 10th from the left and L is 15th from the right. If there are 30 students in total, how many students are between K and L?

- (A) 3 (B) 5 (C) 7 (D) 4

Q8. Four friends J, K, L, M sit around a square table facing center. J is to the left of K. M is to the right of K. Who is opposite to K?

- (A) J (B) L (C) M (D) Cannot be determined

Q9. In a row of 25 girls, Priya is 12th from the left. What is her position from the right?

- (A) 13th (B) 14th (C) 12th (D) 15th

Q10. Five cars Red, Blue, Green, Yellow, White are parked in a row. Red is next to Blue. Yellow is next to White. Green is parked alone in the center. If Red is at the extreme left, where is Blue?

- (A) Extreme right (B) 2nd from left (C) 4th from left (D) Center

Q11. Six friends A, B, C, D, E, F sit in a circle facing center. A is second to the right of B. C is immediate left of B. D is opposite to A. Who is between D and E?

- (A) F (B) C (C) B (D) None

Q12. In a row of soldiers, Amit is 8th from the front and Sumit is 13th from behind. If 4 soldiers are between them, find the total number of soldiers.

- (A) 25 (B) 24 (C) 26 (D) 23

Q13. Four persons E, F, G, H sit in a circle facing outwards. E is to the right of F. G is opposite to F. Who is to the left of G?

- (A) E (B) H (C) F (D) Cannot be determined

Q14. In a straight row of 5 items, Pen is 2nd from left. Pencil is 4th from left. Eraser is between Pen and Pencil. What is the position of Eraser from left?

- (A) 1st (B) 3rd (C) 5th (D) 2nd

Q15. Seven persons sit in a row facing North. X is in the exact middle. 3 persons sit to his left. How many persons sit to his right?

- (A) 2 (B) 3 (C) 4 (D) 1

Q16. Four persons W, X, Y, Z sit around a circular table. W and X are opposite. Y and Z are opposite. If W is to the immediate left of Y, who is to the immediate right of Y?

- (A) X (B) Z (C) W (D) None

Q17. In a row of children, Rohit is 15th from left and Ankit is 7th from right. They interchange their positions, and Rohit becomes 20th from left. Find total children.

- (A) 26 (B) 27 (C) 28 (D) 25

Q18. Five houses P, Q, R, S, T are in a row. P is right of Q. R is left of Q and right of T. S is right of P. Which house is in the middle?

- (A) P (B) Q (C) R (D) T

Q19. Three persons A, B, C sit in a circle facing center. A is left of B. B is left of C. Who is right of A?

- (A) B (B) C (C) Both A and B (D) None

Q20. In a row of 40 persons, Manoj is 13th from the left. What is his rank from the right end?

- (A) 27th (B) 28th (C) 29th (D) 26th

Q21. Four players 1, 2, 3, 4 sit at corners of a table. Player 1 is opposite Player 3. Player 2 is to the left of Player 1. Who is opposite to Player 2?

- (A) Player 1 (B) Player 3 (C) Player 4 (D) None

Q22. In a line of trees, Mango is 5th from start and Neem is 11th from end. If 3 trees are between them, find total trees.

- (A) 19 (B) 18 (C) 20 (D) 17

Q23. Five persons L, M, N, O, P sit in a row. N is center. L and P are at ends. M is left of N. Who is right of N?

- (A) L (B) M (C) P (D) Cannot be determined

Q24. Six desks are arranged in a circular ring. Desk 1 is opposite Desk 4. Desk 2 is adjacent to 1 and 3. Where is Desk 5 located relative to 4?

- (A) Adjacent to 4 (B) Opposite to 2 (C) Opposite to 5 (D) None

Q25. In a row of people, Sunil is 20th from left and Vinay is 15th from right. If they interchange positions, Vinay becomes 30th from right. Find total people.

- (A) 49 (B) 50 (C) 48 (D) 51

Answer Key for Practice Questions

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
Q1	(C) D	Q8	(B) L	Q15	(B) 3	Q22	(A) 19
Q2	(B) Q	Q9	(B) 14th	Q16	(B) Z	Q23	(C) P
Q3	(A) 6th	Q10	(B) 2nd from left	Q17	(B) 27	Q24	(A) Adjacent to 4
Q4	(B) R	Q11	(A) F	Q18	(C) R	Q25	(A) 49
Q5	(B) Chemistry	Q12	(A) 25	Q19	(B) C	-	-
Q6	(B) Y	Q13	(B) H	Q20	(B) 28th	-	-
Q7	(A) 3	Q14	(B) 3rd	Q21	(C) Player 4	-	-